

Fraction mixed operations + four mixed operations

Unit three · Three fraction mix + four first after order · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 3 + Modern Education 5L B Unit 12

Core traps: ☐ T6 Wrong order of mixed operations · T9 Fractional operations - the numerator forgets to synchronize after the common division · The result is not reduced

SSPA association:  High frequency Paper 1 accounts for about 15-20%, Paper 2 accounts for about 10%

Prerequisite knowledge: Course 7-8 (Addition and subtraction of different denominators) · Course 9 (Multiplication of fractions) · Course 22 (Division of fractions)

Goal of this course: ❶ Continuous addition and subtraction of three fractions ❷ Mixed multiplication and division of three fractions ❸ Four rules of order ❹ Comprehensive formula trap

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 4 question · 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate LCM(3, 4, 6) = ? (least common multiple of three numbers)	Basic	
2	Calculation: $\frac{1}{2} + \frac{1}{3} = ?$ (Review addition of common fractions with different denominators)	Basic	
3	Calculation: $\frac{2}{3} \times \frac{1}{2} = ?$ (Review multiplication of fractions)	Basic	
4	Calculation: $\frac{3}{4} \div \frac{1}{2} = ?$ (Review fraction division: the fraction after $\div$ needs to be reversed!)	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Adding and subtracting mixed three fractions ● SSPA

- ① Add and subtract three or more fractions with different denominators → **First find the LCM of all denominators**
- ② Each fraction divides to the same denominator (the numerator is expanded simultaneously)
- ③ The denominator remains unchanged, and the numerator is calculated in the order of addition and subtraction
- ④ The result is reduced to the simplest fraction (improper fraction → mixed fraction)
- ⑤ **Note: LCM needs to find all denominators, not just two!**

Example 1

calculate: $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = ?$

Example 2

calculate: $\frac{3}{4} - \frac{1}{2} + \frac{1}{8} = ?$

Knowledge Point 1 Synchronization practice (must write out: ①Find the LCM that has/have denominator ②find a common denominator ③numerator addsubtract in order ④simplify)

#	Question	Difficulty	Working Space
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5	$\frac{1}{4} + \frac{1}{2} + \frac{1}{8} = ?$		
6	$\frac{2}{3} + \frac{1}{6} + \frac{1}{2} = ?$		
7	$\frac{5}{6} - \frac{1}{3} + \frac{1}{2} = ?$		
8	$\frac{1}{2} + \frac{1}{3} - \frac{1}{4} = ?$		

⚠ Warning: When adding and subtracting three fractions, the LCM must include all denominators. For example, the denominator is 2, 3, 4 → LCM(2,3,4)=12. If it is not LCM(2,3)=6, stop! 4 does not divide 6.

Knowledge point 2: Mixed multiplication and division of third fractions

SSPA required test

Core rule: all division signs become multiplication signs (reverse divisors), then all numerators are multiplied ÷ all denominators are multiplied

- ① When encountering $\div \rightarrow$, reverse the fraction after $\div$, $\div$ becomes $\times$
- ② All multiplications are calculated from left to right
- ③ You can **cross reduction** (The common factor of any numerator can be reduced by any denominator)
- ④ Finally, the numerator is multiplied by the denominator ÷ the denominator is multiplied $\rightarrow$ reduction

Original formula $\div$ Change $\times$, reverse the fraction

$$\frac{1}{3} \times \frac{1}{2} \div \frac{1}{6} \rightarrow \frac{1}{3} \times \frac{1}{2} \times \frac{6}{1} = \frac{6}{6} = 1$$

$\div \frac{1}{6}$
 (divide by a fraction) |||SEP|||

$\times \frac{6}{1}$ (倒數)
 (乘以倒數)

Mnemonic : 「見除變乘 · Frac

❑ Trap detonation example 3 (the most important demonstration in this class)

calculate: $\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4} = ?$

Common mistakes (70% students)

$$\frac{1}{2} \times \frac{2}{3} \div \frac{1}{4} = \frac{2}{6} \div \frac{1}{4} = \frac{2}{6} \times \frac{1}{4} \quad \text{✗}$$

Calculate the previous multiplication first, then forget to reverse when dividing, and mistakenly change $\times \frac{1}{2}$ into $\times 2$ and then $\times \frac{1}{4}$

Correct solution

$$\frac{1}{2} \times \frac{2}{3} \times \frac{4}{1} = \frac{8}{6} = \frac{4}{3} = 1 \frac{1}{3}$$

$\div \frac{1}{4} \rightarrow \times 4/1$, all $\div$ become $\times$, and then reduce from left to right

 **Tip: "Change all division signs into multiplication signs, and reverse the subsequent fractions; cross reduction is the most labor-saving, and the numerator and denominator should be equal."**

Example 4

calculate: $\frac{3}{4} \div \frac{1}{2} \times \frac{2}{3} = ?$

Knowledge Point 2 Synchronization practice (write out: ① $\div \rightarrow \times$ reverse ②crosssimplify ③numerator $\times \div$ denominator $\times$ ④simplify)

#	Question	Difficulty	Working Space
9	$\frac{1}{3} \times \frac{1}{2} \div \frac{1}{6} = ?$		
10	$\frac{2}{5} \div \frac{2}{3} \times \frac{1}{2} = ?$		
11	$\frac{3}{4} \times \frac{2}{3} \div \frac{1}{2} = ?$		
12	$\frac{5}{6} \div \frac{2}{3} \times \frac{1}{2} = ?$		

 **High-frequency errors:** When mixing multiplication and division, if you do multiplication first and then division, you may make an error. The best strategy: first turn all $\div$ into $\times$ (reverse the divisors) and process them uniformly.

Knowledge Point 3: Four Mixed Operation Sequences SSPA Required Exam

Operation precedence (from high to low):

- ① **Parentheses ()**— the calculations inside the parentheses are evaluated first
- ② **Formula: parentheses first, then multiplication and division, and finally addition and subtraction**(from left to right at the same level)
- ③ **+ and -**— Addition and subtraction last (from left to right)
- ④ **Formula: brackets first, then multiplication and division, and finally addition and subtraction**(Same level from left to right)

①

②

③

④

Parentheses first The formula within () must be calculated first	Multiplication and division take precedence $\times \div$ is calculated before + -	Addition and subtraction last $+$ - after parentheses and multiplication and division	Same level left to right Operations at the same level are performed from left to right
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❏ **Trap detonation example 5**

calculate: $\frac{1}{2} + \frac{1}{3} \times \frac{1}{4} = ?$

✗ **Common Mistake — Doing it Directly from Left to Right**

$$\left(\frac{1}{2} + \frac{1}{3}\right) \times \frac{1}{4} = \frac{5}{6} \times \frac{1}{4} = \frac{5}{24}$$

Treat + as if it were on the same level as

✓ **Correct solution: first $\times$ then +**

$$\frac{1}{2} + \frac{1}{12} = \frac{7}{12}$$

Calculate first $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$, then add $\frac{1}{2} = \frac{6}{12} \rightarrow \frac{7}{12}$

Example 6

Calculation: $\left(\frac{1}{2} + \frac{1}{3}\right) \div \frac{1}{6} = ?$ (Note: Calculate first in parentheses, then $\div$)

Knowledge Point 3 Synchronization practice (must mark the calculate order: ①brackets $\rightarrow$ ② $\times \div$ $\rightarrow$ ③ $+-$)

#	Question	Difficulty	Working Space
13	$\frac{1}{2} + \frac{1}{3} \div \frac{1}{6} = ?$		
14	$\frac{2}{3} \times \left(\frac{1}{2} + \frac{1}{4}\right) = ?$		
15	$\frac{3}{4} - \frac{1}{2} \times \frac{1}{3} = ?$		



Tip: "Put parentheses first, then multiplication and division, and finally addition and subtraction; the same level is from left to right, and the order cannot be messed up."

Knowledge Point 4: Comprehensive Application - Four Mixed Text Questions

● **SSPA Required**

Key to solving the problem: ① Understand the question $\rightarrow$ Determine which operations are used ② Column comprehensive calculation (note the brackets and order)

③ Calculate step by step in the four order ④ Write a complete answer(Step points will be deducted if you do not write an answer!)

Example 7

Xiao Ming has $\frac{1}{2}$ cakes, and his mother buys twice as many $\frac{1}{3}$ cakes. What fraction of the cake does Xiao Ming have now?

Example 8

A rope is $\frac{3}{4}$ meters long. After using $\frac{1}{2}$ meters, the remaining $\frac{2}{3}$ of the rope is used for handicrafts. How much rope did you use to make it?

Knowledge Point Four Synchronization Practice

#	Question	Difficulty	Working Space
16	Xiaomei has $\frac{2}{5}$ liters of orange juice. After drinking $\frac{1}{10}$ liters, she divides the remainder into 3 glasses. How many liters are there in each cup?		
17	One piece of $\frac{1}{3}$ was used as a gift, and the remaining $\frac{1}{2}$ was used for decoration. What fraction of the total rope does the decorative part account for?		
18	The younger brother's pocket money belongs to the older brother $\frac{1}{3}$, and the younger brother's pocket money belongs to the younger brother $\frac{1}{4}$. What fraction of my sister's pocket money is that of my brother's? $\frac{1}{2}$, the sister's is the brother's $\frac{2}{3}$. What fraction of my sister's pocket money is that of my brother's?		

III. Lesson Layered Synchronization Practice

Basic layer (total 4 questions, everyone must do)

#	Question	Difficulty	Working Space
19	$\frac{1}{3} + \frac{1}{6} + \frac{1}{2} = ?$		

20	$\frac{5}{8} - \frac{1}{4} + \frac{1}{2} = ?$		
21	$\frac{1}{4} \times \frac{2}{3} \div \frac{1}{2} = ?$		
22	$\frac{1}{2} + \frac{1}{4} \div \frac{1}{2} = ?$		

Advanced layer (total 3 questions,   choose do)

#	Question	Difficulty	Working Space
23	$\frac{3}{5} + \frac{1}{2} - \frac{1}{10} + \frac{1}{5} = ?$ (Four fractions addition and subtraction mixed)		
24	$\frac{7}{8} \div \frac{3}{4} \times \frac{1}{2} = ?$		
25	$(\frac{1}{2} - \frac{1}{3}) \div \frac{1}{6} + \frac{1}{4} = ?$		

 challenge layer (total 2 questions,   choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
26	$\frac{1}{2} + \frac{2}{3} \times \frac{3}{4} \div \frac{1}{2} = ?$ (Four mixing rules: first $\times \div$, then $+-$)		
27	$\frac{2}{3} \div \frac{1}{2} \times (\frac{1}{2} + \frac{1}{4}) = ?$ (Parentheses first, then multiplication and division from left to right)		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
1	Calculation: $(\frac{1}{2} + \frac{1}{3}) \times (\frac{2}{3} - \frac{1}{4}) \div \frac{1}{6} = ?$ (Two brackets, then $\times \div$ - pay attention to the order!) $\frac{1}{2} + \frac{1}{3} \times (\frac{2}{3} - \frac{1}{4}) \div \frac{1}{6} = ?$ (Two parentheses, then $\times \div$ - pay attention to the order!)		
2	Bottle of Juice $\frac{3}{4}$ Liters. I drank liters of $\frac{1}{6}$ at breakfast and the remaining $\frac{2}{3}$ at lunch. How many liters are left after drinking lunch? (Calculation using comprehensive formulas)		

Advanced mixed question (👤🚀 choose do)

#	Question	Difficulty	Working Space
M1	$\frac{1}{2} + \frac{1}{3} \div \frac{1}{6} - \frac{1}{4} = ?$ ($\div \rightarrow - \rightarrow +$, pay attention to the order!)		
M2	$\frac{2}{3} + \frac{1}{2} \times (\frac{1}{3} + \frac{1}{6}) = ?$ (Brackets first, then $\times$, and finally $+$)		

V. Class afterhomework

Basic must-doquestions (total 6 questions, mustwrite outcalculate order mark)

#	Question	Difficulty	Working Space
H1	$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} = ?$		
H2	$\frac{3}{5} + \frac{1}{10} + \frac{1}{2} = ?$		
H3	$\frac{2}{3} \times \frac{1}{4} \div \frac{1}{2} = ?$		
H4	$\frac{3}{4} - \frac{1}{8} + \frac{1}{2} = ?$		
H5	$\frac{1}{2} + \frac{1}{3} \times \frac{1}{4} = ?$		
H6	$\frac{3}{5} + \frac{1}{2} \times \frac{2}{3} = ?$		

For advanced, choose doquestion (total 2 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H7	$\frac{2}{3} \div \frac{1}{2} + \frac{1}{4} \times \frac{2}{3} = ?$		
H8	A ribbon is $1\frac{1}{2}$ meter long. After using SEP , the remaining $\frac{1}{3}$ is used to wrap gifts. How much rice did it take to wrap the gift? $\frac{1}{2}$ Used to wrap gifts. How much rice did it take to wrap the gift?		

VI. The Lessoncorecommon errorssummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🍌 basic questions independently

☐ I can challenge 🌿 advanced questions

☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🍌 basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Three fractions LCM only finds two SEP : denominator 2,3,4 only finds LCM(2,3)=6: Denominator 2,3,4 only looks for LCM(2,3)=6	LCM must include all denominators! LCM(2,3,4)=12
2	When mixing multiplication and division, forget to change ÷ into × : $\frac{1}{2} \times \frac{1}{3} \div \frac{1}{6}$ and only count the first two	All ÷ become × (the divisor is reversed) and are processed uniformly
3	Four order errors—adding and subtracting first, then multiplying and dividing	First ×÷ and then +−, if there are parentheses, it is calculated first
4	Uncommon points in brackets : $(\frac{1}{2} + \frac{1}{3})$ Write directly $\frac{2}{5}$	The scores in parentheses must be divided first and then calculated!
5	During cross reduction, only the same denominator group is reduced	The common factors of any numerator can be reduced by any denominator
6	The result is not reduced/improper fraction is not transferred	The final step must be to check the reduction and transfer of mixed fractions
7	The order of expressions in application questions is reflected incorrectly(missing brackets)	Check when formulating: What does the question ask you to do first? Do I need to use parentheses?